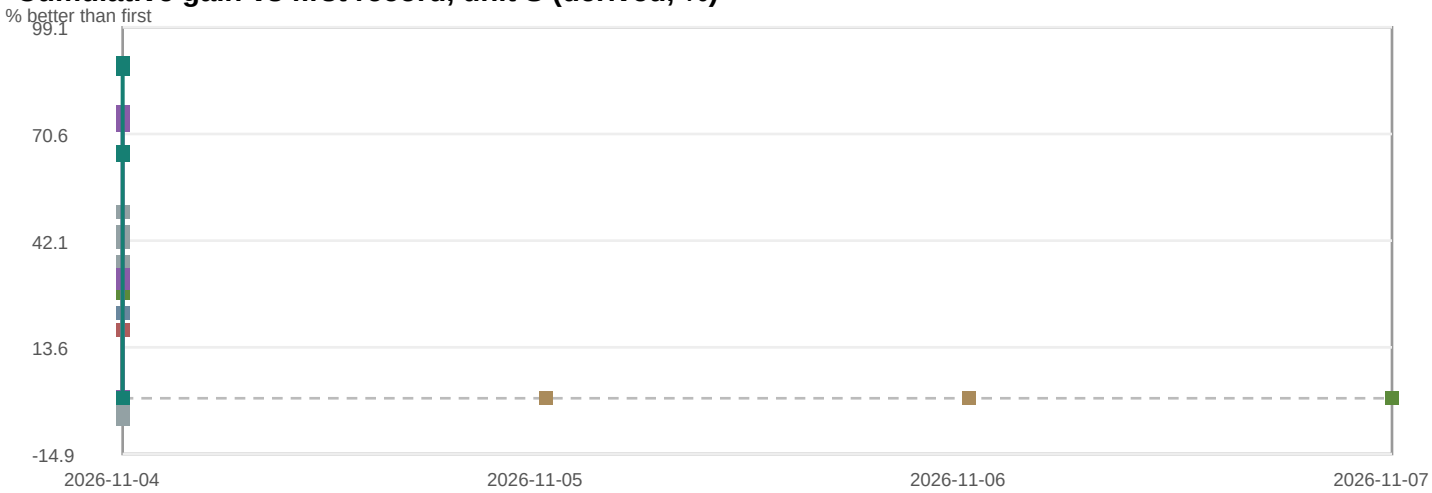
Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 49, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

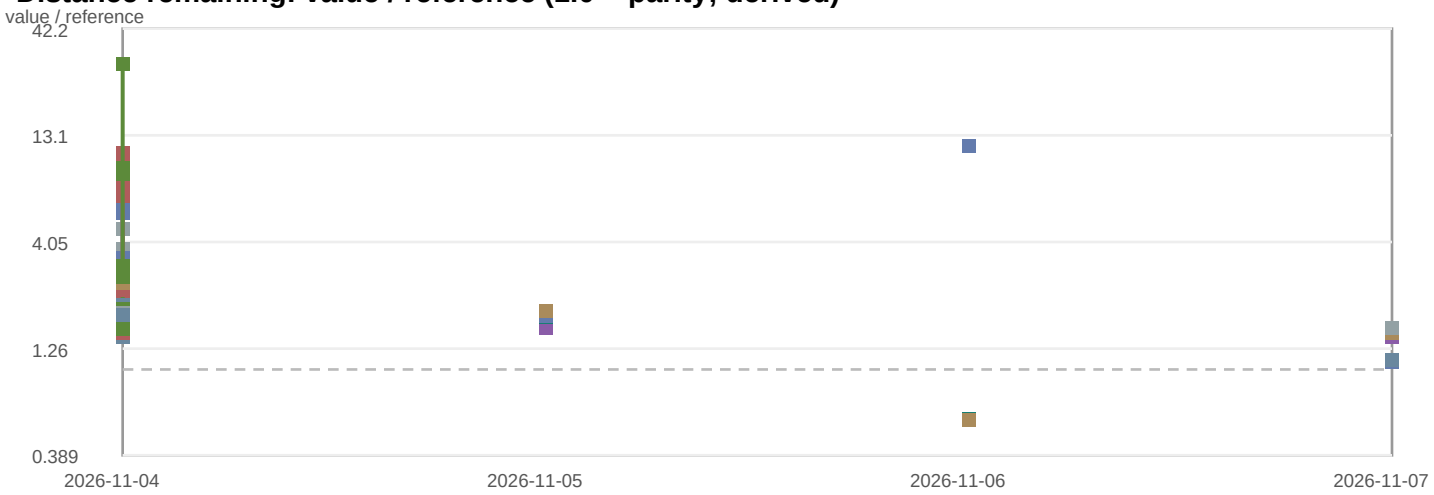
Cumulative gain vs first record, unit S (derived, %)



Series key

- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

Distance remaining: value / reference (1.0 = parity; derived)



Series key

- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
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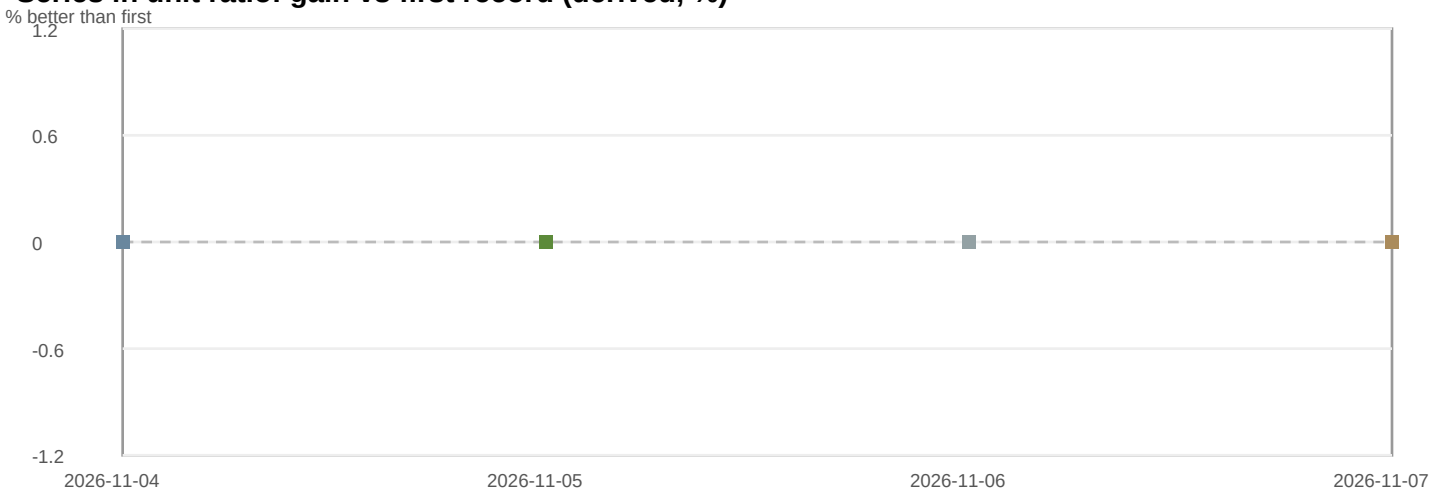
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

Series in unit ratio: gain vs first record (derived, %)



Series key

- analysis:ecbench_statistics_v2
- analysis:ecbench_statistics_v2
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

Series in unit ratio: gain vs first record (derived, %) (series key continued)

Series summary

problem • metric	unit	first	latest	gain vs first	value/ref	pts
analysis:ecbench_statistics_v2 • censored_ratio_	ratio	648.2	648.2	-	-	1
analysis:ecbench_statistics_v2 • heavy_tail_rows	count	71	71	-	-	1
analysis:ecbench_statistics_v2 • ops_interval_wi	ratio	13.1	13.1	-	-	1
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	0.5739	0.5739	0%	0.5814	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1	S	12.2426	12.2426	0%	11.6066	2
dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_	S	0.5819	0.5819	0%	0.5746	2
ic:curve25519-family-k20-32 • ratio_w0_k24	ratio	0.9375	0.9375	0%	-	1
ic:curve25519-family-k20-32 • ratio_w0_k32	ratio	0.8802	0.8802	0%	-	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k20	S	2.5746	2.5746	0%	1.7601	1
ic:curve25519-family-k20-32 • s_w0_incumbent_k24	S	2.1638	1.6735	22.6592%	1.429	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k28	S	1.9211	1.5717	18.1849%	1.4943	2
ic:curve25519-family-k20-32 • s_w0_incumbent_k32	S	2.11	1.5152	28.1881%	1.5539	2
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.601	2.601	0%	1.5669	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	2.111	2.111	0%	1.7695	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.869	1.869	0%	1.7971	1
ic:curve25519-family-k20-32 • s_w0_mean5curves_k	S	1.93	1.93	0%	1.9052	1
ic:curve25519-family-k20 • s_w0_incumbent	S	8.3478	4.8598	41.7827%	3.2356	5
ic:curve25519-family-k24-28 • s_w0_incumbent_k24	S	2.5842	2.5842	0%	2.0304	1
ic:curve25519-family-k24-28 • s_w0_incumbent_k28	S	2.3732	2.3732	0%	2.3865	1
ic:curve25519-family-k24-32 • best_train_s_w0_k2	S	2.713	2.713	0%	-	1
ic:curve25519-family-k24-32 • best_train_s_w0_k3	S	2.768	2.768	0%	-	1
ic:curve25519-family-k24-32 • ratio_w0_k32	ratio	0.9106	0.9106	0%	-	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k24	S	2.1055	2.1055	0%	1.942	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k28	S	1.8835	1.8835	0%	1.8154	1
ic:curve25519-family-k24-32 • s_w0_seedrun_k32	S	1.8423	1.8423	0%	1.8232	1
ic:curve25519-family-k24 • s_w0_incumbent	S	6.607	3.3182	49.7782%	2.7163	5
ic:curve25519-family-k28-32 • s_w0_incumbent	S	2.5798	2.5798	0%	2.5891	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k28	S	1.969	1.969	0%	1.8593	1
ic:curve25519-family-k28-32 • s_w0_seed_mean_k32	S	1.781	1.781	0%	1.8229	1
ic:curve25519-family-k28 • s_w0_incumbent	S	10.9395	2.576	76.452%	2.7572	7
ic:curve25519-family-k32 • s_w0_incumbent	S	24.1377	2.519	89.5641%	2.7453	5
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2283	1.2283	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.2385	1.2385	-0%	-	1
ic:curve25519-k20-single-target • ratio_f2_over_	ratio	1.1542	1.1542	-0%	-	1
ic:curve25519-k20-single-target • s_f1_x07_w01	S	2.3781	2.3781	0%	1.4287	1
ic:curve25519-k20-single-target • s_f1_x07_w0	S	2.3211	2.3211	0%	1.4951	1
ic:curve25519-k20-single-target • s_f1_x07_w1	S	2.8907	2.8907	0%	1.0818	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1475	1.1475	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.1636	1.1636	-0%	-	1
ic:curve25519-k24-single-target • ratio_f2_over_	ratio	1.0384	1.0384	-0%	-	1
ic:curve25519-k24-single-target • s_f1_x07_w01	S	1.8945	1.8945	0%	1.4975	1
ic:curve25519-k24-single-target • s_f1_x07_w0	S	1.8341	1.8341	0%	1.5802	1
ic:curve25519-k24-single-target • s_f1_x07_w1	S	2.4383	2.4383	0%	1.1058	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	0.936	0.936	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	2.883	2.883	-0%	-	1
kernel:inherited_f4_solve_tree • wall_seq_over_p	ratio	1.589	1.589	-0%	-	1
kernel:inherited_f4_solve_tree • work_par_over_s	ratio	2.865	2.865	0%	-	1
registry-rebinding:ic-aimed-gather • raw_pins_re	count	211	211	-	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	0.718	0.718	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0948	1.0948	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.4723	1.4723	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0653	1.0653	-0%	-	1
rho:koblitz_signed_frobenius_panel • s_legacy_ov	ratio	1.0008	1.0008	-0%	-	1

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared. 6 digest-pinned historical measured records are excluded from all series and derived comparisons; recorded status counts retain them. See integrity_exclusions in progress.json and PROGRESS.md for reasons.

Historical records excluded from comparisons

benchmarks/records/20261105T0828Z-icb-large-prime-n53-s1-archive.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.

benchmarks/records/20261105T0828Z-icb-large-prime-n53-s2-archive.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.

benchmarks/records/20261105T0828Z-icb-large-prime-n61-s1-archive.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.

benchmarks/records/20261105T0828Z-icb-large-prime-n61-s2-archive.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.

benchmarks/records/20261105T1759Z-ic-packed-filter-p32.json: Documentary retrospective commands have no captured successful exit status. Values remain historical quotations; execution provenance is incomplete.

benchmarks/records/20261105T2109Z-f6-few-candidates.json: Incomplete spread endpoints: the historical metric has a lower endpoint and null upper endpoint; it cannot satisfy the finite two-endpoint measurement contract.